

NMAS – Chartmetric Historical Extraction Brief (Developer Implementation Spec)

Objective

Generate historical reference-period datasets required by NBS for YoY and QoQ analysis. NBS requires that we state in detail where and how each variable is gotten, what constitutes the variable, and how it is generated.

Reference Periods (Mandatory)

- **Oct–Dec 2024** (Q4 2024)
- **Jul–Sep 2025** (Q3 2025)
- **Oct–Dec 2025** (Q4 2025)

Data Source Constraint

All “market/streaming” variables to be extracted from Chartmetric Developer API historical endpoints. Where Chartmetric does not provide a metric, geography, or granularity, we will:

- Deliver the field as “Not available” (or omit it), and
- Document the limitation clearly (no interpolation or inference) for subsequent phases or developments.

1) Scope

1.1 Entity Scope (Required)

- **Primary entity:** Nigerian artists (Chartmetric artist_id)
- **Secondary entity:** track (Chartmetric track_id) - required because Chartmetric’s strongest consumption time series (e.g., Spotify streams, YouTube views) are often track-led.

Entity universe input: a list of:

- artist_name + (preferred) chartmetric_artist_id
- Where possible, top_track_ids per artist or sufficient metadata for Chartmetric resolution.

1.2 Geographic Scope (Required)

- **Primary geo:** Nigeria
- **Secondary geo:** Global (for export calculation)

Important: If an endpoint does not support geo filtering, developers must label the output field geo_scope="global" and flag the limitation in methodology notes.

1.3 Frequency / Granularity (Required)

- **Preferred:** daily time series per metric/entity
- **Acceptable:** weekly time series

NMAS will compute quarterly aggregates internally; therefore, daily/weekly is preferred for auditability.

2) Output Format

2.1 Required Files

1. **Fact table(s):** nmas_chartmetric_metrics.csv (or partitioned by entity_type)
2. **Entity dimension tables:** nmas_artists.csv, nmas_tracks.csv
3. **Methodology note:** nmas_variable_methodology.md or .docx
4. **Extraction notes:** nmas_extraction_notes.md (endpoints, limitations, coverage)

2.2 Fact Table Required Columns

- date
- period_label (Q4_2024 / Q3_2025 / Q4_2025)
- entity_type (artist/track)
- entity_id (Chartmetric ID)
- entity_name
- platform (as supported by Chartmetric; e.g., Spotify, YouTube, Pandora, Shazam).
- geo_scope (nigeria/global)
- variable_name (NMAS-defined)
- variable_value
- unit
- source_endpoint (Chartmetric endpoint name/path used)
- source_field (field name within response)
- extraction_timestamp

3) Variables

A) Core Consumption Variables

Platform_streams/views/counts_daily (Track-level) (Platform name as supported by Chartmetric, e.g., Spotify / YouTube / Pandora / Shazam)

- **Where obtained:** Chartmetric tracks historical “stats/trends” for platform streams/views/counts.
- **What constitutes it:** Daily count of platform streams/views/counts for the track.

- **How generated:** Pull daily series for each track for dates covering the three reference periods; store daily values; NMAS will aggregate totals within each period.
- **Unit:** streams/views/counts.

B) Location and Reach Proxy Variables

Platform_audience_reach_daily (Artist-level, if available) (Platform name as supported by Chartmetric)

- **Where obtained:** Chartmetric platform artist stats time series.
- **What constitutes it:** Location and reach indicators reported per day (as stored by Chartmetric), including (where available) monthly listeners and “where people listen” distribution by city/country over time.
- **How generated:** Pull time series across periods; store daily/weekly values as available. Extract city/country-level series if present; otherwise, document limitation.
- **Unit:** listeners (or share), depending on the field.

4) Period Aggregation Rules (For NBS Tables)

NMAS will compute the quarterly reference-period tables as:

- **Period total (volume metrics):** sum of daily values within period
- **Net change (reach proxies):** last observation minus first observation within period

5) Methodology Notes (Mandatory for NBS)

For each variable_name, developers must provide a short methodology entry stating:

- **Source:** Chartmetric Developer API
- **Endpoint used:** (exact path/name)
- **Field used:** (exact response field)
- **Definition:** what the metric represents in Chartmetric terms
- **Unit:** streams/views/count, etc.
- **Generation:** extraction + any AI computation performed (e.g., sum, average, net change)
- **Coverage limitations:** geo constraints, missing dates, platform exclusions, entity coverage gaps, etc.